

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> The power dissipated by the bulb is given by $(P = IV)$. The power also equals $(P = \frac{V^2}{R})$, where (V) is the voltage and (R) is the resistance of the bulb . We are given that $(P = 40)$ watts and the voltage $(V = 12)$ volts . Let's calculate the resistance of the bulb :

$$[40 = \frac{12^2}{R}]$$

Solving for (R) :

$$[40 = \frac{144}{R}]$$

$$[R = \frac{144}{40} = 3.6 \, , \, \Omega]$$

Now , we use the power equation to find the current (I) :

$$[P = I \times V]$$

$$[40 = I \times 12]$$

$$[I = \frac{40}{12} = 3.33 \, , \, \text{A}]$$

So , the current is 3.33 amps and the resistance is 3.6 ohms . </think>

<answer>E </answer>